

Matrices

4.1 Matrices

Outline

1. Types of matrices, name, and properties
2. Different operations of matrices (Addition and Multiplication)
3. Eigen values and Eigenvectors
4. Functional Matrices: Diagonalization and Similarity transformation

Definition: Matrices are rectangular arrays of numbers or functions.

Example: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ or $A = \begin{bmatrix} e^x & \sin x \\ x & e^{-x^2} \end{bmatrix}$

Applications:

- It can carry a very large amount of data.

- Scientific and Mathematical ideas can be expressed explicitly.
- Handy for storing and processing the data.
- Very useful in Quantum Mechanics

Representation

$$\begin{bmatrix} & \\ & \end{bmatrix}_{m \times n} \Rightarrow m = \text{Rows}, n = \text{Columns}$$

$$\begin{bmatrix} a_{11} & b_{12} \\ c_{21} & d_{22} \end{bmatrix}_{2 \times 2} \Rightarrow \text{Rows} = 2, \text{Columns} = 2$$